

Infrastructure for the stochastic chart assembly

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2026-09-07

Abstract

Unit 214 (Astra #23 programme G, step 1) provides the probabilistic glue for assembling finitely many charts with jointly random phases: a deterministic convergent sequence converges in measure; finitely many sequences converging to 0 in measure give a vector converging to 0 (sup norm); Slutsky pairing of a sequence converging in distribution with a deterministic convergent sequence (a wrapper around Mathlib's `TendstoInDistribution.prodMk_of_tendstoInMeasure_const`); and the *joint vector form of Headline XVI*: for charts $a \in \kappa$ (finite) with their own dimensions and exponents, jointly convergent random phases $X_n \Rightarrow Z$ in $\prod_a C([0, 1]^{d_a+1})$ and fixed amplitudes g_a , the vector $(F_{N_n}^{(a)}(X_n^a, g_a))_a$ converges in distribution to $(F^{(a)}(Z^a, g_a))_a$ (`tendstoInDistribution_normChart_vec`). Zero sorry/axiom; 9091 jobs.

1 The things formalised

```
theorem tendstoInMeasure_const_of_tendsto (c :  $\alpha \rightarrow E$ ) (c) (hc : Tendsto c L (c)) :
  TendstoInMeasure (fun n _ => c n) L (fun _ => c)
theorem tendstoInMeasure_pi_zero [Fintype  $\alpha$ ] (f :  $\alpha \rightarrow \Omega \rightarrow E$ )
  (hf :  $\forall a, \text{TendstoInMeasure } (f a) L (\text{fun } _ => 0)$ ) :
  TendstoInMeasure (fun n a => f a n) L (fun _ => (0 :  $\Omega \rightarrow E$ ))
theorem tendstoInDistribution_prodMk_tendsto_const (hX : TendstoInDistribution X L
  Z (fun _ => ) ' )
  (c :  $\alpha \rightarrow E'$ ) (c) (hc : Tendsto c L (c)) :
  TendstoInDistribution (fun n => (X n , c n)) L (fun _ => (Z , c)) (fun _
    => ) '
abbrev PhaseSpaces (d :  $\mathbb{N}$ ) := (a :  $\mathbb{N}$ )  $\rightarrow$  C(closedCube (d a + 1),  $\mathbb{R}$ )
theorem tendstoInDistribution_normChart_vec ... (X :  $\mathbb{N} \rightarrow \Omega \rightarrow \text{PhaseSpaces } d$ ) (hXm)
  (Z) (hZm) (hX) (Nseq) (hN)
  (hN0 :  $\forall n, 0 \leq Nseq n$ ) (g : (a :  $\mathbb{N}$ )  $\rightarrow$  C(closedCube (d a + 1),  $\mathbb{R}$ )) :
  TendstoInDistribution (fun n a => normChart (h a) (k a) (l a) (Nseq n) (X n
    a, g a)) L
  (fun a => limChart (h a) (k a) (l a) (Z a, g a)) (fun _ => ) '

```

2 Mathematics

Nothing new analytically: the vector statement is Headline XVI componentwise, with the errors collected by the union bound (`measure_iUnion_fintype_le` behind `pi_norm_lt_iff`) and the limits by continuous mapping of the phase vector. The deterministic-coefficient Slutsky lemma is what will turn the per-chart normalised integrals into the assembled sums $\sum_a I_a/S_* = \sum_a (S_a/S_*) (I_a/S_a)$ with $S_a/S_* \rightarrow e_a \in \{0, 1\}$.

3 Lean notes

- Mathlib's `prodMk_of_tendstoInMeasure_const` needs the first space to be a `SeminormedAddCommGroup` with `BorelSpace` and `SecondCountableTopology`; a wrapper stated with a bare `TopologicalSpace` does not typecheck.
- Composing `continuous_apply a` inside `continuous_comp` leaves the `Pi` type metavariable unresolved; state the composite continuity as a typed `have` first.
- The Borel σ -algebra on the factors gives the product σ -algebra on `PhaseSpaces` `d` for free (`measurable_pi_iff`, `measurable_pi_apply`).

4 Next

Unit 215: Headline XVIII, the assembled stochastic posterior quotient: the pair vector $(F_a^{\mathcal{P}^c}, F_a^c)_a$, Slutsky with the deterministic scale ratios, continuous summation, a.s.-positive selected denominator, and the exact cancellation of the common scale for $N_n > 1$.